

19. (a) Describe the energy transfers taking place in a wind generator. [2]

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(b) The maximum power that can be extracted from a wind generator is given by  $P = \frac{1}{2} \rho A v^3$ .  
State **one** assumption made in deriving this equation. [1]

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(c) The following data are available for a wind generator:

- Blade radius = 2.5 m
- Air density = 1.2 kg m<sup>-3</sup>
- Wind speed incident on blades = 6.8 m s<sup>-1</sup>
- Wind speed after leaving blades = 2.6 m s<sup>-1</sup>

Determine the maximum power, stating the correct unit, produced by this wind generator. [2]

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